Example

- How many ways are there to select a first-prize winner, a second-prize winner, and a third-prize winner from 100 different people who have entered a contest?

- **Solution:**
  The order matters as it matters which person wins which prize. Therefore, the number of ways to pick three prize winners is the number of *ordered* selections of three elements from a set of 100 elements, i.e., the number of 3-permutations of a set of 100 elements.

$$P(100,3) = 100 \times 99 \times 98 = 970200 \text{ ways.}$$